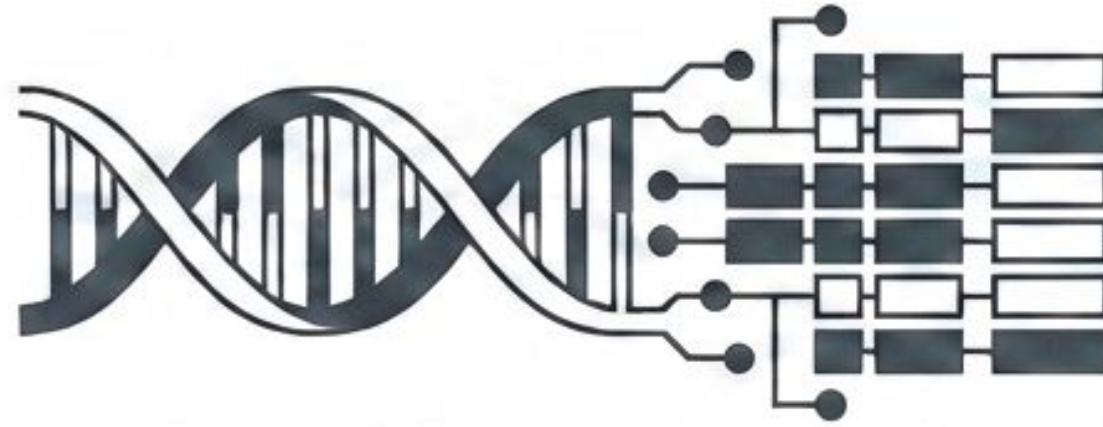
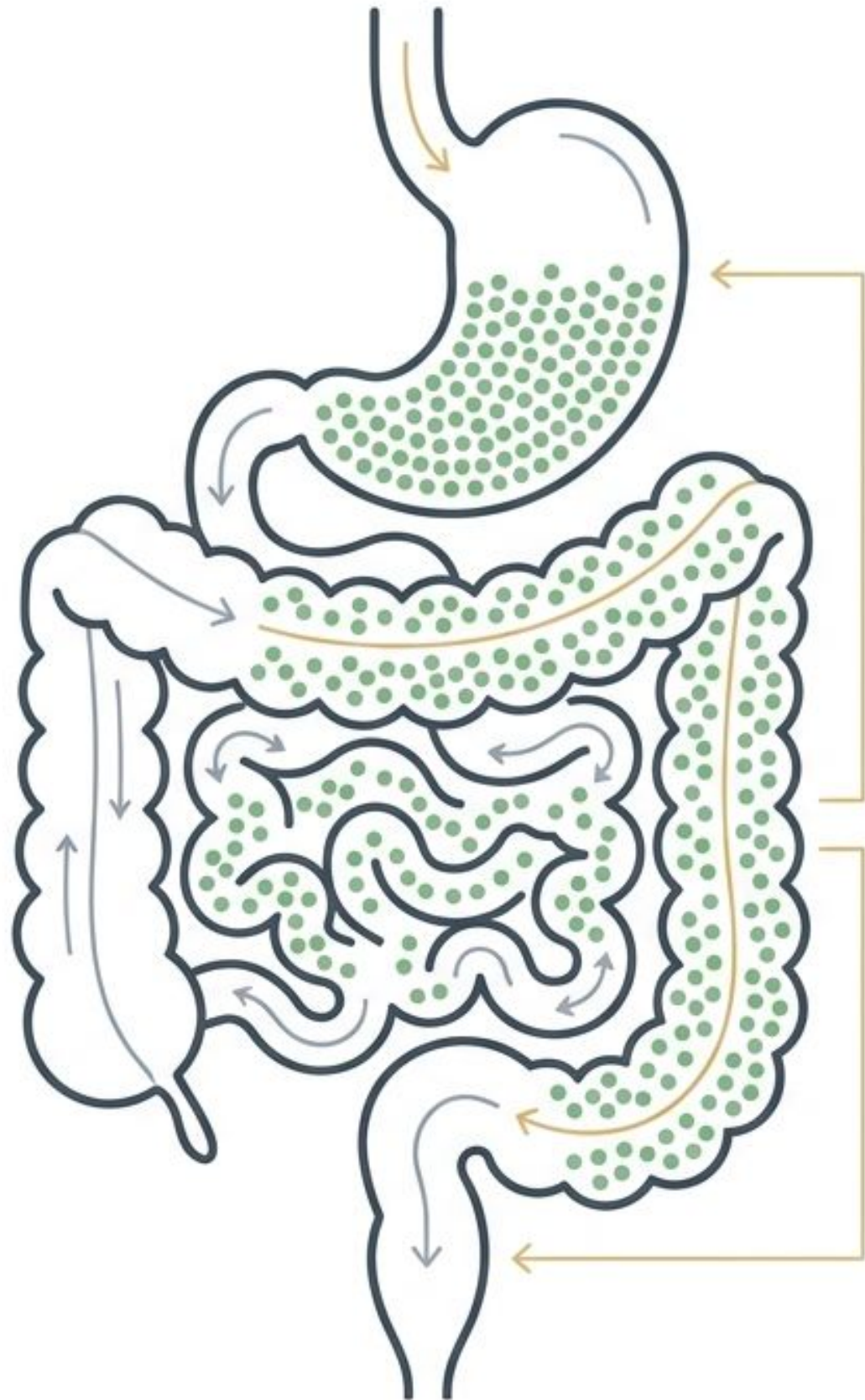




Colorectal Cancer Prediction via Microbiome Analysis using Machine Learning

(A BCD Analytics Hub Workshop Project)

**A Comprehensive Training Workshop
Project Learnings
BUSHRA KHAN**



The Biological Challenge

The human gut microbiome interacts heavily with host immunity. Our objective is to determine if we can diagnose Colorectal Cancer (CRC) non-invasively by analyzing these distinct bacterial communities.

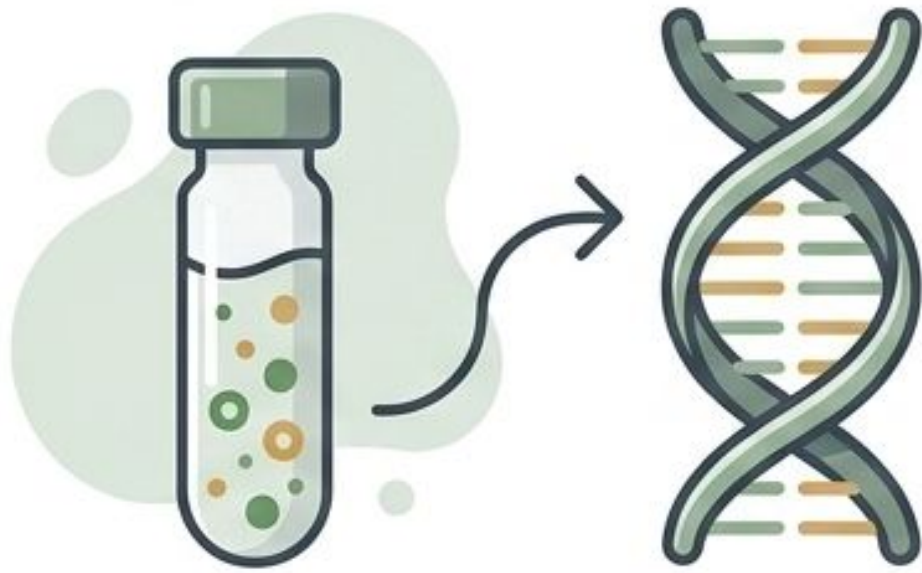
The Computational Approach

Digitally process raw stool metagenomes, algorithmically profile the exact bacterial taxonomy, and deploy Machine Learning to mathematically classify patient states.

Target Output

Discovery of predictive microbial signatures that quantitatively discriminate healthy controls from CRC patients.

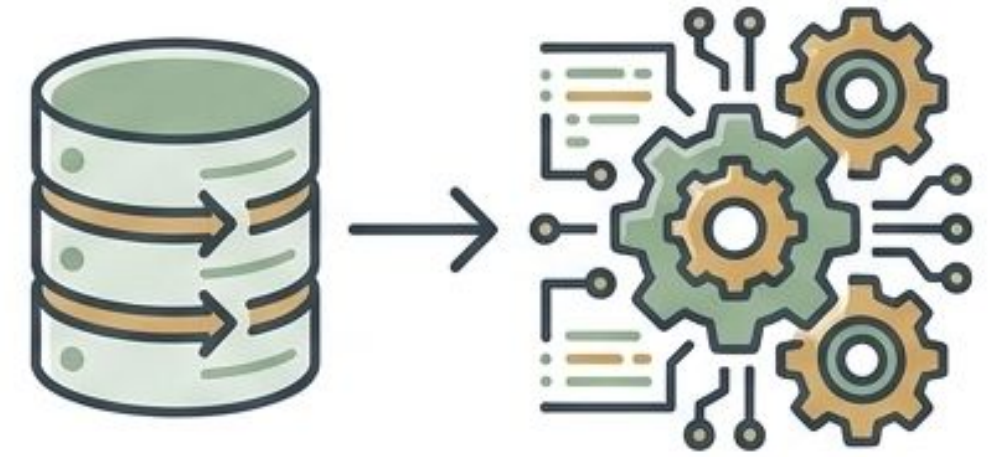
Wet Lab (Physical)



- Fecal sample collection and DNA extraction

Next-Generation Sequencing (NGS) Digitization

Dry Lab (Computational)

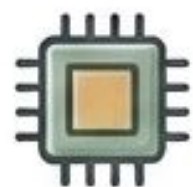


- Algorithmic processing, Machine Learning feature extraction, and clinical intelligence

Methodology Rationale: Why Shotgun Metagenomics?

Unlike 16S sequencing which targets a single identifier gene, Shotgun Metagenomics shears and sequences all genomic DNA randomly. This achieves the absolute species and strain-level resolution mandated by machine learning algorithms to build robust predictive features.

Computational Environment



macOS (Native Unix Terminal) |
Miniconda Virtual Environments

Constraint: 8GB RAM Minimum due to
intensive genomic alignment overhead.

Dataset Scale



1,545 total patient samples

Aggregated from 6 distinct BioProjects
(e.g., PRJNA1237248, PRJEB72524)

Metadata Integration (Pandas)

Sequencing IDs
(SraRunTable)

Clinical Diagnoses
(META_file)

Sequencing IDs
(SraRunTable)



Clinical Diagnoses
(META_file)

Machine learning requires a clinical 'Source of Truth'. We utilized Pandas in Python to programmatically link sequence data to clinical records generating accurate **Target (Y) labels: CRC Stage, Tumor Location, and Diagnosis.**



NCBI / ENA
Sequence Read Archive



HTTP Transfer



Local Linux Storage

Secure Download (prefetch)

The SRA Toolkit rapidly downloads highly compressed binary biological data (.sra files).



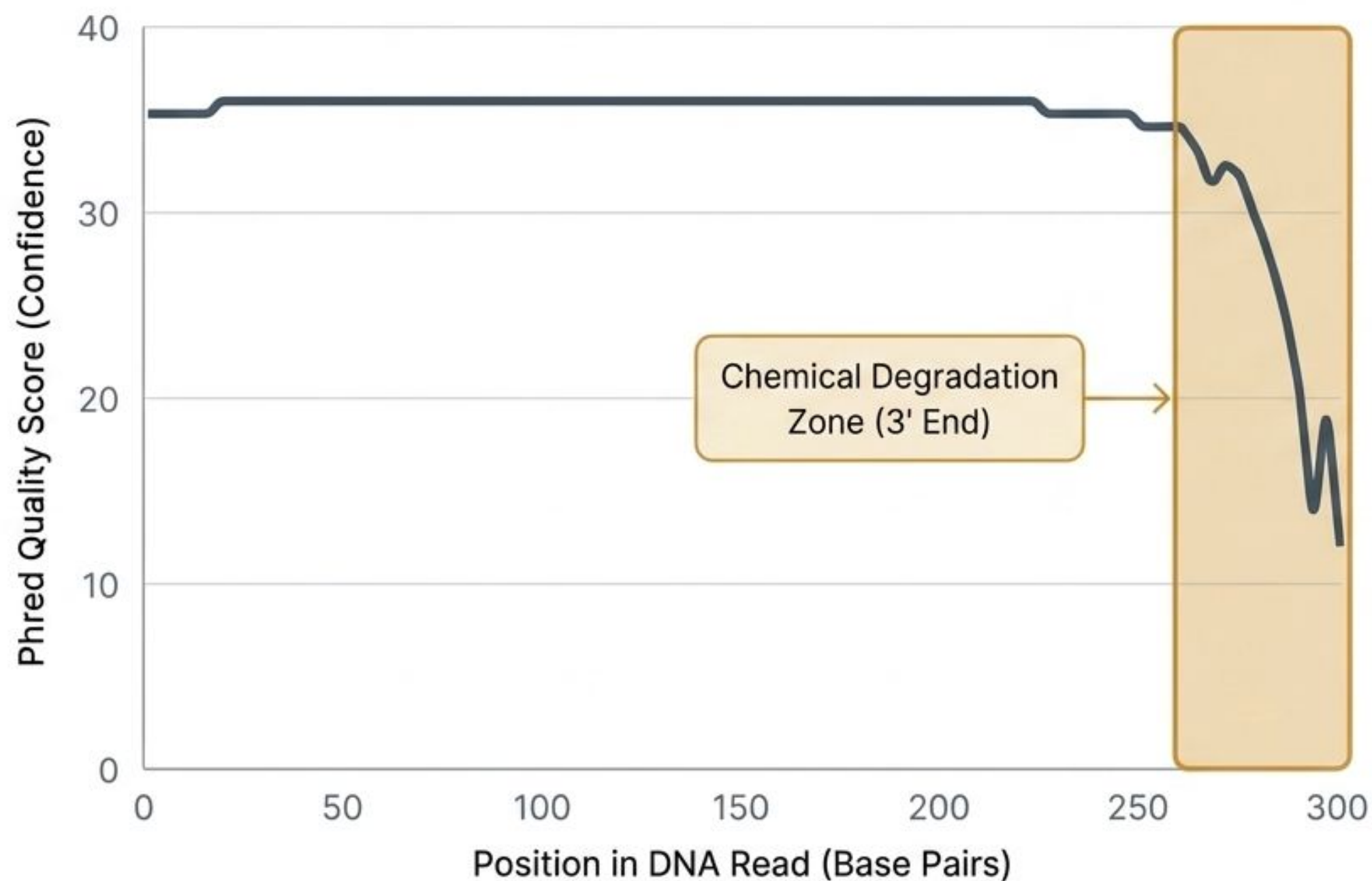
ASCII Extraction (fastq-dump)

The binaries are unpacked into readable sequence text.

Biological Requirement: Spatial Synchronization

The `--split-files` flag is biologically crucial. It forces the algorithmic toolkit to separate paired-end reads into distinct **Forward (R1)** and **Reverse (R2)** files maintaining the 3D spatial synchronization required for downstream alignment.

Quality Control: The Degradation Graph



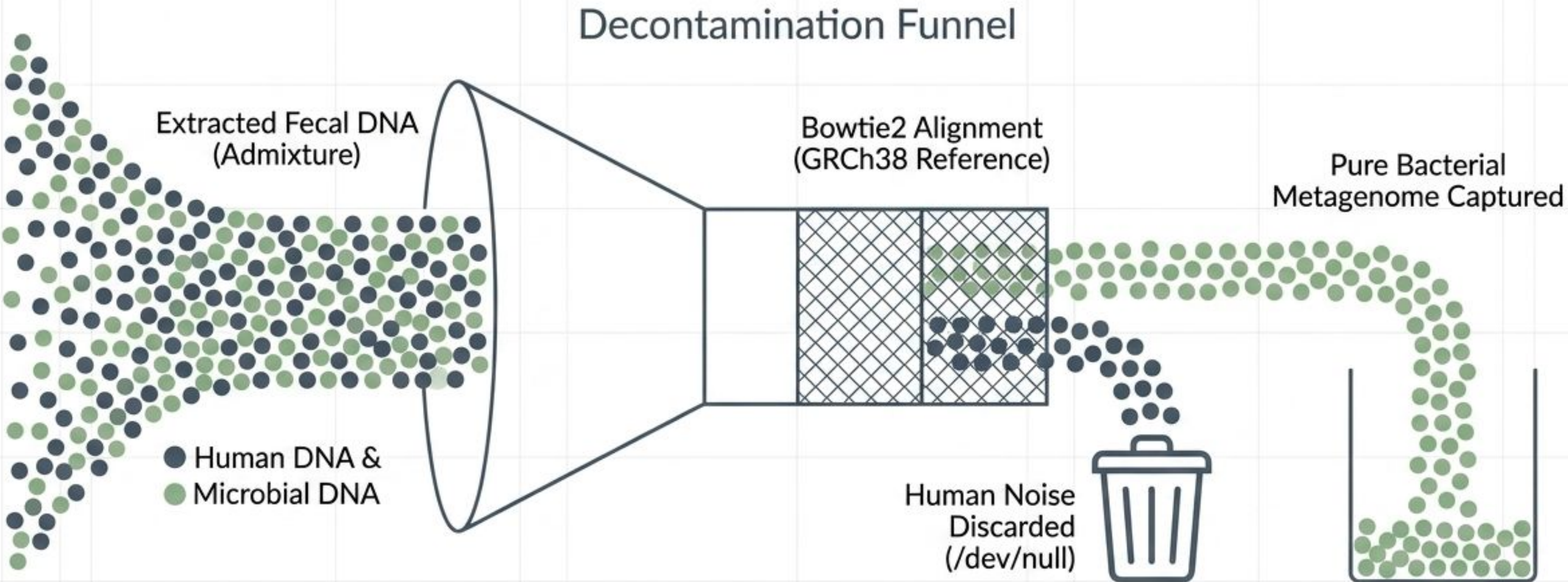
The Problem

Chemical sequencing cycles induce physical error accumulation. Base call reliability rapidly decreases at the trailing 3' ends of DNA reads.

The Solution (fastp)

We deployed fastp as a computational scalpel mathematically enforcing data integrity by purging segments falling below a Q30 threshold (99.9% accuracy). This prevents machine learning models from learning false mutations.

Host Decontamination: Filtering Human Noise



Sample ERR14218891			
1.	Total Input Reads:	22.4 Million	
2.	Failed to map to Human Genome:	99.99%	
3.	Result:	Successful isolation of microbial signal from human host noise.	

Taxonomic Profiling: Choosing an Approach

Option 1: Local Execution (Kraken2 & MGnify) - SELECTED	Option 2: Cloud-based (MetaPhlAn 4 & Zenodo)
	
Goal: Identify species using exact-match k-mer queries against a gut database. Result: Feasible. Requires significant computational resources (RAM) for large databases but provides high-resolution, exact-match profiling.	Goal: Acquire finalized, normalized species abundance tables. Result: Feasible. Bypasses local hardware limits by utilizing pre-computed taxonomic profiles hosted in the cloud.
Reason for Selection: Kraken2 & MGnify chosen for its high precision and exact-match capability, critical for distinguishing between closely related microbial strains in high-dimensional datasets.	

Feature Space Engineering: Structuring the Data

1 The Feature Matrix (X)

A unified grid containing the numerical abundances of every identified microbe across all verified patients.

Patients
(1 to 1,545)

Features: Normalized Microbial Species (X)							Target Vector: Clinical Truth (Y)
0.045	0.112	0.238	0.003	0.567	0.890	...	1
0.049	0.132	0.340	0.313	0.883	0.345	...	0
0.003	0.067	0.062	0.017	0.302	0.480	...	1
0.038	0.008	0.015	0.149	0.713	0.560	...	1
0.003	0.034	0.023	0.339	0.527	0.270	...	0
0.006	0.082	0.037	0.103	0.765	0.165	...	0
0.007	0.003	0.036	0.018	0.809	0.452	...	0
...	...	...	...	...	...	...	0...

2 The Target Vector (Y)

Binary encoding of the clinical diagnosis where CRC = 1 and Healthy = 0.

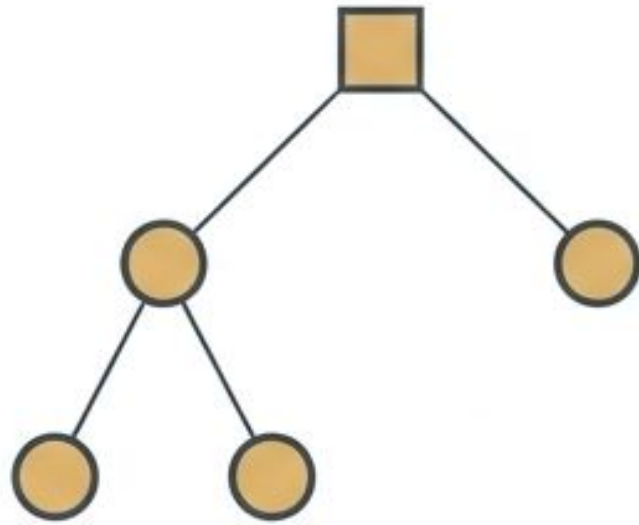
3 Sanitization Protocol

Rows missing clinical truth (NaN) were systematically dropped to prevent algorithmic corruption.

Algorithm Selection: Managing High Dimensionality

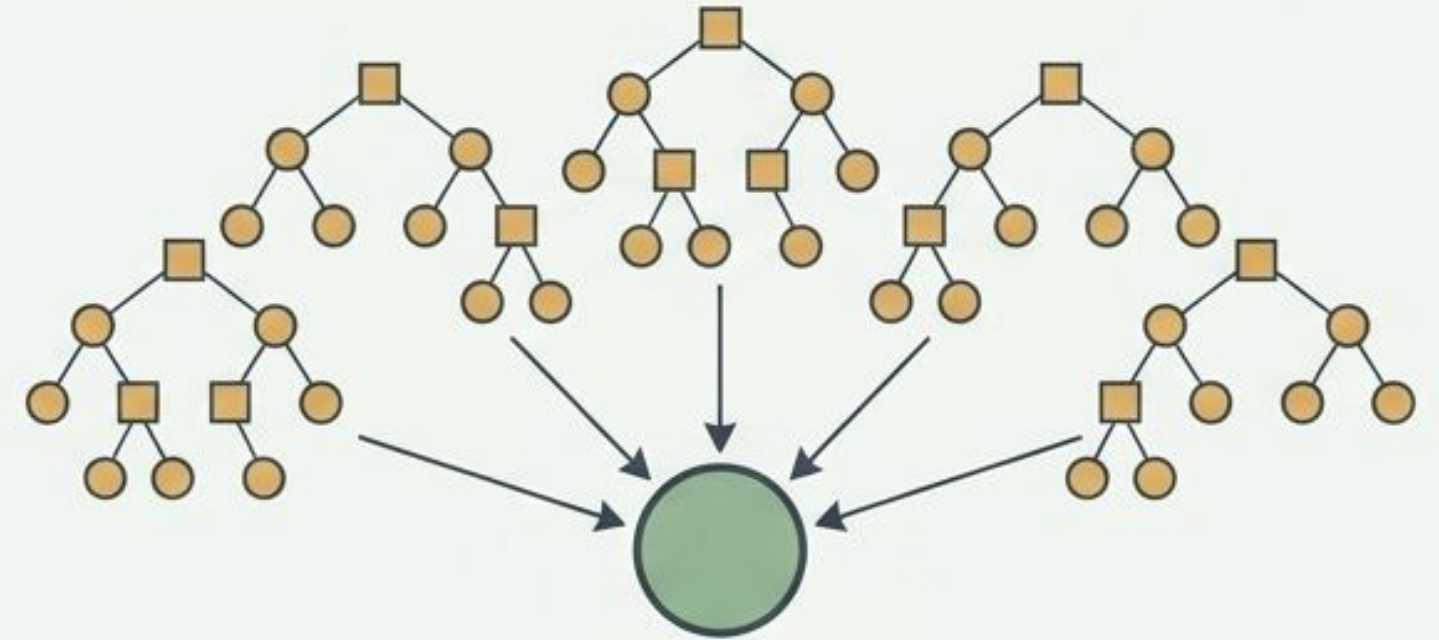
The Data Reality: Microbiome datasets are inherently sparse and highly dimensional (more features than samples) with intense non-linear relationships.

Single Decision Tree (Rejected)



Prone to memorizing noise. A deep tree drastically overfits the training data while a shallow tree underfits failing to capture complex biological distinctions.

Random Forest Classifier (Selected)



An ensemble method that averages hundreds of distinct decision trees. It naturally manages high dimensionality, resists dataset overfitting, and negates the need for strict mathematical scaling.

Validation Protocol: Preventing Memorization

100% Core Dataset

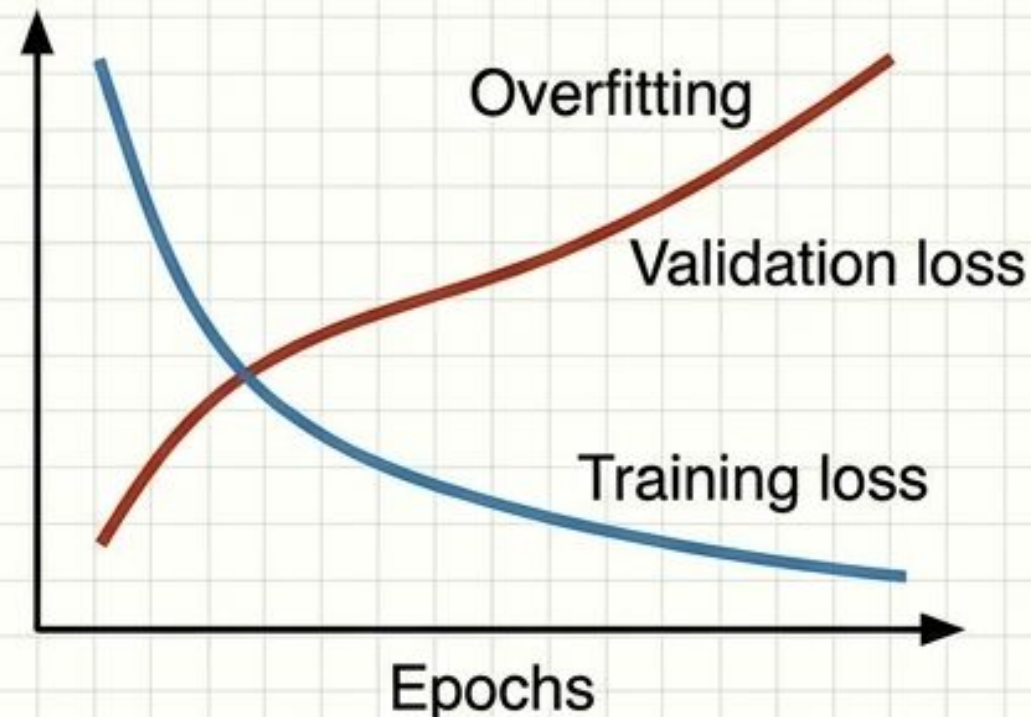
Training Data: For fitting the algorithm



Testing Data: Reserved for final validation

The Overfitting Trap

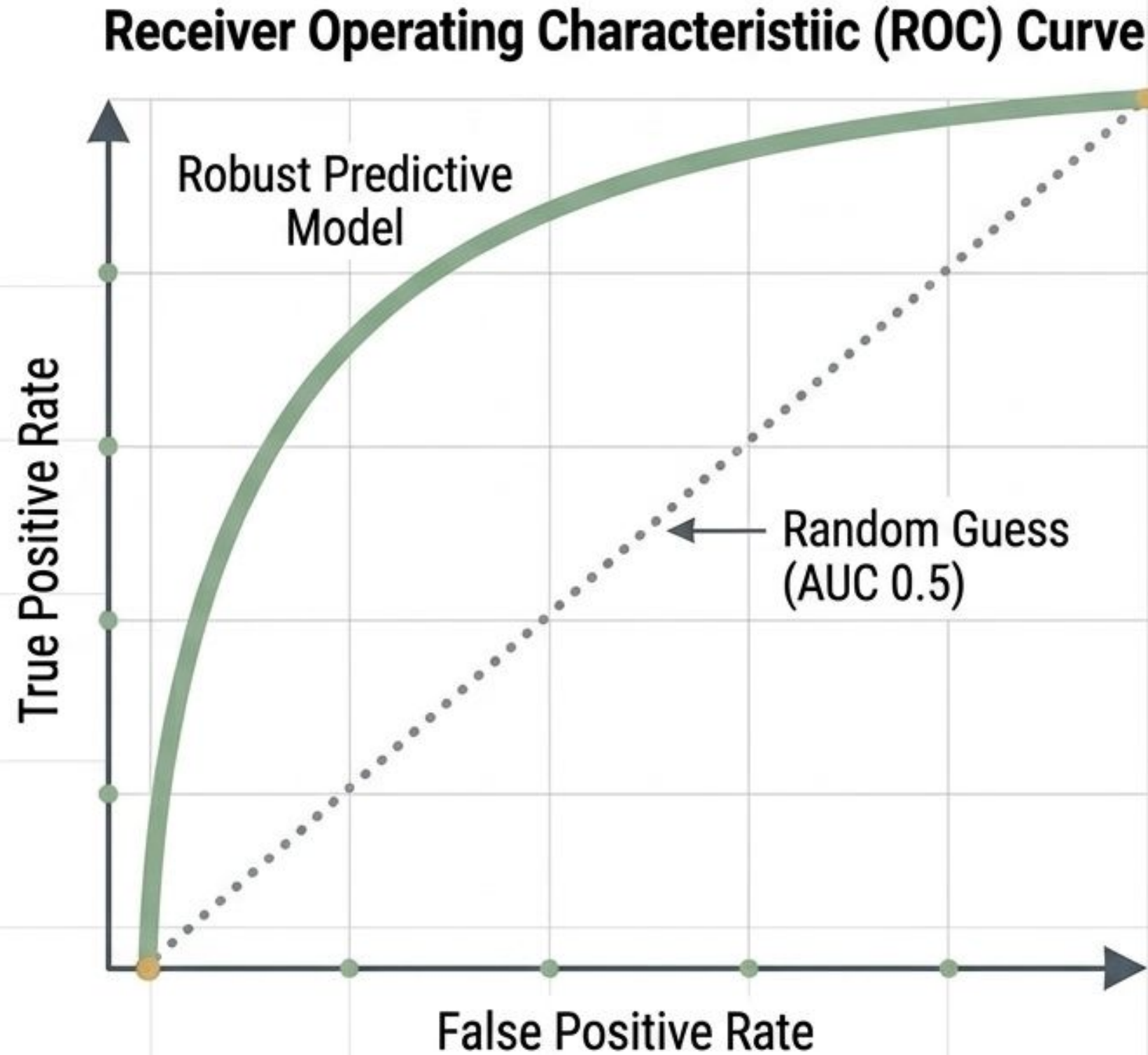
Testing a model on its own training data results in perfect but entirely false accuracy. The model memorizes the data rather than learning the underlying biological rules.



Parameter Tuning

The `random_state` variable is manually tuned to evaluate stability against algorithmic randomness proving the model has learned generalizable patterns.

Model Evaluation: The Accuracy Trap in Medical Data



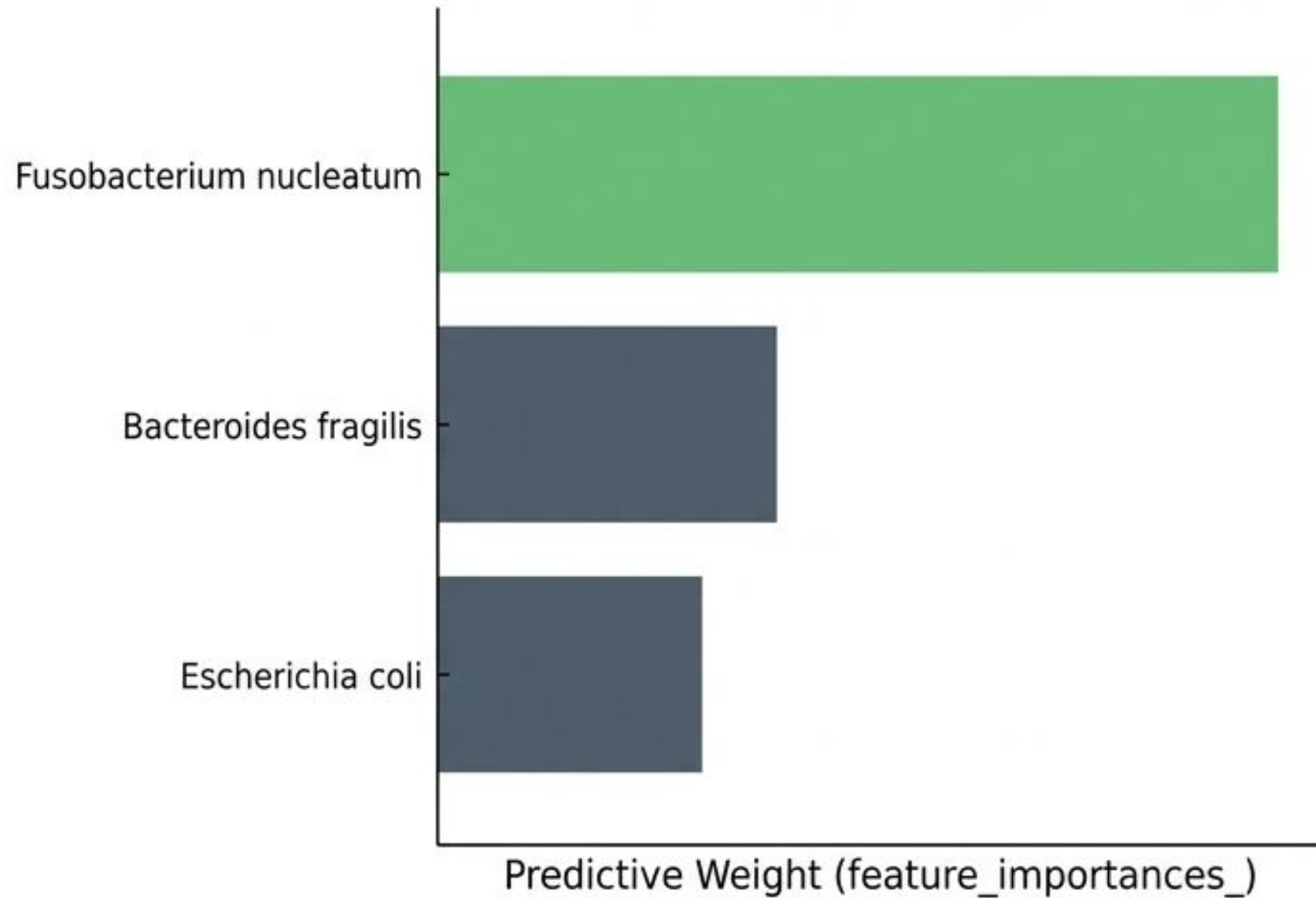
The Flaw of Basic Accuracy

In imbalanced medical data (e.g., 90% healthy, 10% sick), a model that blindly guesses 'Healthy' every time is mathematically 90% accurate but clinically useless.

Clinical Viability (AUC-ROC)

We evaluate performance using the Area Under the Curve (AUC-ROC) which tests every threshold to define the model's absolute capability to distinguish a CRC patient from a healthy control mitigating dataset bias.

Biomarker Discovery: Opening the Black Box



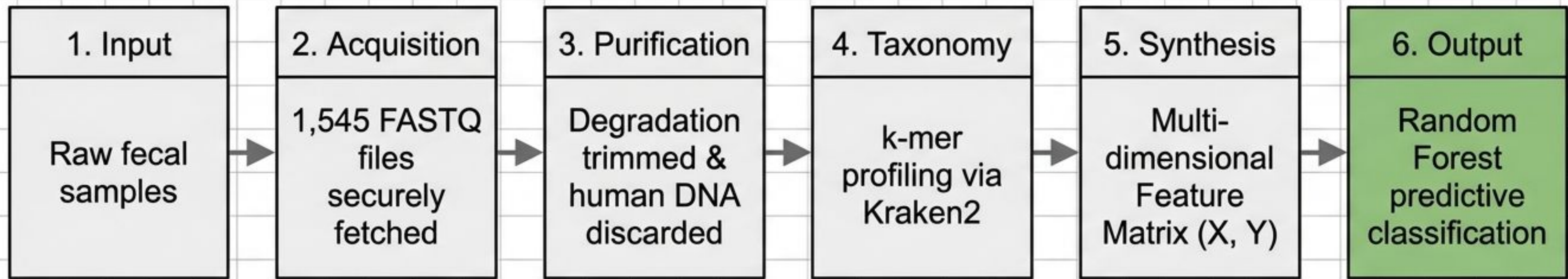
Algorithmic Transparency

Unlike opaque neural networks, Random Forests allow for the direct extraction of mathematical logic. The algorithm ranks which specific variables most heavily dictated node splitting across its decision trees.

Clinical Interpretation

The top-ranked features mathematically correlate to the biological taxa driving the prediction. By identifying reliance on microbes like *Fusobacterium nucleatum*, the model effectively acts as an automated discovery engine for cancer pathogenesis.

The Data Refinery: End-to-End Pipeline Architecture



Conclusion: Pipeline Results & Interpretation

Data Types Processed	Results Achieved	Interpretation & Mastery
 Successfully transitioned physical biology across three digital states: Raw sequencing reads (ASCII FASTQ files) into processed taxonomic profiles (Normalized Numerical Tables) and finally into clinical Machine Learning targets (Binary Vectors).	 The Random Forest successfully distinguished CRC samples from healthy controls. By properly structuring validation splits we circumvented overfitting to extract a stable model capable of automatically identifying pathogenic markers like <i>Fusobacterium nucleatum</i>.	 A highly accurate predictive model is entirely dependent on meticulous upstream dry-lab execution. Managing severe computational resource constraints, manually enforcing data integrity, and navigating parameters demonstrated hands-on mastery of bioinformatics.